AI APPROACHES IN PERSONALIZED MEAL PLANNING FOR A MULTI CRITERIA PROBLEM

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The Supervisory Committee certifies that this ***disquisition*** complies with North Dakota State University's regulations and meets the accepted standards for the degree of

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ABSTRACT

Food is one of the necessities of life. The food we consume every day provides us with the nutrition we need to have energy. However, food plays a more significant role in life. There is a relationship between food, culture, family, and society [1]. Since ancient civilization, people have realized the correlation between food and healthiness. Earlier, Physicians were treating people by prescribing special recipes. Last century, assorted studies investigated the impact meals have on human nutritional intake and the different diseases connected to it. There have been numerous other studies that focused on the required nutritional intake to ensure a good amount of energy for well-being in humans. A person who advises individuals on their food and nutrition is known as a dietarian and nutritionist. Nowadays nutritionists are experts in the use of food and nutrition to promote health and manage disease. They suggest several diet rules and food recommendations to assist people in living a healthy life.

Due to technological advancements, previous time-consuming issues that required human attention are now being solved by utilizing automated procedures machines. Meal planning is one of the attractive domains that recently has received great notation by researchers who are using machine learning techniques in it. In general, those studies were performed to use extracted nutrition knowledge and food information for designing an automated meal planning system. However, in the majority of published research, the user's preferences were an ignored feature.

In this research, my journey through developing automated meal planning systems unfolds across distinct projects, each building upon the insights and advancements of its predecessors. Starting with a focus on incorporating user preferences, the exploration evolved through successive iterations, seeking to mirror the complexities of real-world decision-making more accurately. This progression led to the integration of advanced methodologies spanning artificial intelligence,

optimization, multi-criteria decision making, and fuzzy logic. The ultimate aim was to refine and enhance the systems to not only align with users' dietary restrictions and preferences but also to adapt to user feedback, thereby continually improving their efficacy and personalization. Through this comprehensive approach, the research endeavors to contribute novel solutions to the nuanced challenges of personalized meal planning.

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1. UNDERSTANDING THE PROBLEM

1.1. Introduction

"You are what you eat," goes the old saying, highlighting the profound impact of our dietary choices on our health and well-being. This simple yet profound statement underscores the importance of nutrition in our lives, serving as a prelude to a deeper exploration of the complex relationship between food and health. To further our understanding, we delve deeper into the issue, examining how dietary habits contribute to the prevalence of chronic diseases and exploring various methods and aspects that play a crucial role in meal planning.

1.2. Problem

Food serves as a cornerstone in people's lives, impacting both personal well-being and social interactions. Beyond mere sustenance, it significantly influences health, fitness, and medical treatment. Numerous studies underscore the prevalence of chronic diseases in the United States, and unfortunately, this trend is expected to worsen across all age groups in the coming years [2, 3]. Chronic diseases often stem from a range of risky behaviors, with poor nutrition topping the list [3, 4]. Research consistently links inadequate dietary choices to serious health problems such as obesity, diabetes, and other related issues [5, 6]. On the other hand, rich nutrition plays a meaningful role not just in disease prevention and a healthy lifestyle, but also in the treatment and management of chronic diseases [5, 6].

By ensuring a balanced intake of carbohydrates, proteins, and fats, meal planning helps regulate blood glucose levels for individuals with diabetes. For those with heart disease, meal planning becomes even more critical. A heart-healthy diet—one low in saturated fats and abundant in fruits, vegetables, and whole grains—can mitigate the risk of high blood pressure and elevated cholesterol levels. Beyond prevention, meal planning aids weight management and adherence to medication schedules. By avoiding overeating and aligning meals with treatment regimens, individuals with chronic diseases can improve their overall health outcomes.

In light of all these aspects, how can people decide on the right food? It is important to select a healthy meal that satisfies their health and body needs, as well as their preferences, culture, and limitations.

In terms of health, there are various guidelines and rules that help determine to develop a balanced meal [7-9]. Many recommendations made by nutritionists and dietitians to develop a balanced meal is a complex task. In order to design an ideal diet plan, a number of factors must be taken into account, including nutrition rules, culture, food restrictions, preferences, costs, and the proper composition to get optimum nutrition in a day. Individuals must put in time and effort to read menu terms, compose optimum meals, prepare a weekly schedule, and organize groceries [10]. As a result, people reach out to nutritional experts to help them achieve a healthy lifestyle. However, a few barriers come to play when using a nutritional expert. For example, the cost of visiting an expert. Individuals must meet with an expert regularly to discuss their nutritional and dietary intake. A situation or status may change abruptly, which is difficult for human experts to respond to immediately, and the user receives the updated plan.

There has been a lot of research conducted in the past decades to aid people in making decisions regarding this ongoing and important need for meal planning by using artificial intelligence and designing applications. A popular alternative approach is using online resources such as mobile applications and websites that are highly available and provide essential health and nutrition services. (e.g., eatright, MyPlate, FDA, Myfitnesspal, Lifesum, CalorieCounter, Fooducate) [11-18]. Most of these e-solutions can be categorize to two main group of the calorie tracking applications and meal planning applications. However, they share many common weaknesses in these solutions including (i) shortage of a complete automated process either requiring expert involvement, e.g., [19-21], or involving the patient (user) to have some technical knowledge about nutritional health in order to utilize (and tune the parameters of) the e-solution, e.g., [22, 23], (ii) performing meal planning or meal plan evaluation based on solely predefined diet or nutrition rules, e.g., [19-21], and (iii) permitting limited adaptability to the patient's preferences in terms of diet group, e.g.,[16, 24, 25]. Hence, there is a need for an automated solution to produce a meal plan from scratch, based on user restrictions and a recommended caloric intake and considering multiple patient preference factors.

The main goal of this study is to create an intelligent system that offers a meal planning service that considers specific users' preferences, other health conditions, cultures, and traditions. We aim to automate the process of producing personalized meal plans allowing patients to reach their required daily nutrition intake while closely catering to their preferences. Solving this problem with all these certain and uncertain variables in multi factor solution space, makes this problem to be a NP-hard problem.

In the following chapters, we will delve into the various solutions we have developed to address the complex problem of personalized meal planning. Each solution is designed with a focus on different aspects of the challenge, reflecting our comprehensive approach to creating an effective and user-friendly dietary planning tool. From leveraging advanced artificial intelligence algorithms to incorporating multi-criteria decision-making frameworks, we aim to cover a broad spectrum of needs and preferences. Our exploration includes the development of innovative applications that not only consider nutritional guidelines and health conditions but also embrace user-specific factors such as cultural preferences, budget constraints, and lifestyle choices. Through these chapters, we will detail the methodologies employed, the technological advancements harnessed, and the outcomes achieved, offering insight into how each solution contributes to simplifying the meal planning process for individuals seeking to improve their dietary habits and health outcomes.

1.3. Related Works

This section briefly discusses articles related to computerized solutions connecting to nutritional health and meal planning. Numerous research studies in meal planning provide us with details on the purpose of their objective, data, computational techniques, and other factors. Based off the articles, we investigate the related works in two different aspects. First, we examine the different perspectives on meal planning and, second, the techniques used to solve the meal planning problem. As illustrated in Table 1, we offer a summary of various methods and techniques employed for addressing the meal planning issue by different studies, providing a concise overview of related works in this field.

1.3.1. Various Perspectives

Generally, we can discern different goals pursued by the researchers. We analyzed numerous research and their main purpose on how to solve meal planning problems. We classified these works in the following groups: calorie/nutrition tracker, optimization of meal recommender for special age group, and personalize meal plans. There are articles that correlate with different groups presented.

1.3.1.1. Calorie/Nutrition Tracker

Some studies were focused on designing a meal planner which considers calories or nutrition needed in a day. In [26], a computer-based method for menu planning has been introduced. The algorithm plans three main meals per day for n-days. They decomposed the planning problem into several subproblems at the daily menu and meal-planning level. Then the problem is reduced to a multi-dimensional knapsack problem and feasible solutions are obtained through an evolutionary algorithm, the Elitist Non-Dominated Sorting Genetic Algorithm.

1.3.1.2. Special Age Groups

There is research to optimize healthy nutrition recommendations for users with special constraints. For instance, Hazman and Idrees proposed a prototype expert system for children's nutrition [27]. It generates healthy meals for children of different ages according to different criteria including their growth stage, gender, and their health status.

There is work analyzing the diet of elderly users, for example, the work proposed by Espín, Hurtado, and Noguera [28]. They present "Nutrition for Elder Care," which intends to help elderly users to draw up their own healthy diet plans following the nutritional expert's guidelines. However, they do not provide a real dish in their work.

1.3.1.3. Personalized Meal Plans

Some studies tried to consider user preferences but generally, flavor and diet were considered as preferences. Similar systems include the recommender system proposed by Ribeiro et al [29]. It creates a personalized meal plan based on the information provided by the elderly user, including the anthropometric measures, personal preferences, and activity level. Ribeiro et al. propose a solution for assisting older adults with the planning of meals and shopping, by offering personalized meal recommendations that integrate with external food provisioning services for the delivery of products [30]. Yang et al. analyzed the limitations of existing meal recommendation systems such as the coarse-grained elicited preferences and cold start problem [31]. A personalized nutrient-based meal recommender system is proposed based on individuals' nutritional expectations, dietary restrictions, and fine-grained food preferences via a visual quiz-based user interface. Nikahat et al. [32] propose a system that asks over some personal information and uses fuzzy logic to find variables such as BMI and required daily calorie intake. They also ask some questions to find user preference type diet. Mary in [33] focuses more on user preferences and use fuzzy logic to find the best meals based on different constraints such as budget, cooking skill, variety, time available for cooking, and organization of freshly prepared or pre-prepared.

1.3.2. Different Techniques

The following section will briefly discuss the different techniques utilized in the research studies.

1.3.2.1. Linear Optimization

A smart diet consultant was introduced in [34]. The diet consulting problem was modeled as a mathematical multi-objective optimization problem, and it also accepts feedback from users to fine-tune their meal plans. Kuo et al. proposed a graph-based algorithm to solve the menu planning problem [35]. It uses online cooking recipes associated with ingredients and cooking procedures. Users can specify ingredients. Recipes that contain the query ingredients are returned. First, the proposed approach constructs a recipe graph to capture the co-occurrence relationships between recipes from the collection of menus. Then, a menu is generated by the approximate Steiner Tree Algorithm on the constructed recipe graph. In their work, Elsweiler et al. tried to achieve a balance between healthier food and tastier food [36].

1.3.2.2. Meta- Heuristic

Meta-heuristic is a high-level concept used to find and create the most sufficient answers to a NP-hard problem. Within this concept there are different techniques to be used to optimize the solutions. One of the most well-known algorithms is known as PSO. Particle swarm optimization is used in many studies to improve the result in np-hard problems. In [37] they used PSO to minimize the cost of meals and select them for the user. Or in [38] PSO is used to solve the problem with different aspects such as food variety, calorie intake and satisfying different nutrition.

1.3.2.3. Cased Based Reasoning

Case based reasoning is the process of tackling new issues based on discoveries of past problems. During [19, 20] recommendations for target patients, they select and adapt the best meal plan from an existing one based on the patient's nutritional needs, then revise it accordingly. In [39] they took the existing diet plans into the system and used Case-based Reasoning to propose diet plans. Afterward, he examines the system to filter out irrelevant cases and selects the most appropriate case to suggest to the patient based on Rule-based Reasoning.

1.3.2.4. Fuzzy Reasoning Logic

Fuzzy logic is an approach that helps humans compute more humanly than the machine. Previously, computers worked with Boolean logic known as "true or false" and exact numbers (1 or 0) but fuzzy logic introduces "degrees of truth". Fuzzy logic is used in different parts of meal planning problems. In some research it is used to rank healthiness of existing meal planning systems [40] In some other, it helps to improve efficiency of other approaches in decision process [41]. Kljusurić et. al., [42]use fuzzy logic to analyze and optimize the process of meal planning which is based on the DASH diet. Paul et al. in [43] design a recommender system by fuzzy logic and k-nearest neighbor in a situation where there is no feedback or rating available for training machine learning model.

Salloum in [44] tried to make a novel framework that relies on the fuzzy logic paradigm for central pre-requites to the meal planning task. The fuzzy logic helps to simulate health assessment capabilities, and in their next study [45], they use that framework to design a meal plan. They presented a meal plan composite food primary category. They calculate calories of macronutrients and consider them totally to cover the calorie intake needed and did not consider the effect of each nutrient. This system might not respond properly in all situations, for instance, meal planning for diabetic patients. In this disease, carbohydrate choice per meal should be counted separately.

1.3.2.5. Multi Criteria Decision Making

MCDM techniques help to evaluate multiple conflicting criteria in decision-making. Recently, few different research has been done using different multi criteria decision making approaches for meal planning problems. They mostly help researchers to consider different discrete variables that are important in meal recommendation. In [46] the fuzzy TOPSIS method is applied to investigate qualification factors for preparing the diet of diabetic patient. Toledo et. al, in [47] considered nutrition and preferences synchronized. They use AHP-sort for finding the best meals. Also, they calculate macronutrient calories and such as protein and carbs then find foods with exactly the same calorie amounts needed. They use AHP for filtering and removing foods that are not nutritionally appropriate. Then, use the optimization model and user feedback to find the best options to suggest. In their study, the impact of different nutrition is missed too. While AHP works with pairwise comparison, it cannot be used easily for selecting from a long list of alternatives. So, in their study, they benefit from it just by filtering bad options.

In delving into the complex realm of meal planning, numerous research endeavors have aimed to bridge the delicate balance between adhering to nutritional guidelines, accommodating user preferences, and navigating the practical constraints of daily life. Upon reviewing these studies, it's evident that while some have closely approached the dual objectives of our investigation—proposing meal recommendations that honor both the health aspects and personal preferences of users—these efforts, too, are not without their limitations. This reflection not only highlights the diversity of approaches within the field but also underscores the nuanced challenges that persist in crafting solutions that adeptly merge health considerations with individual dietary desires. Through this analytical lens, we uncover valuable insights and pinpoint areas that are ripe for further innovation and improvement, setting the stage for our contributions to address these identified gaps and propel the domain of personalized meal planning forward. Mary's strategy [33], with its focus on a broad array of considerations including budget, cooking skills, and time availability, employs fuzzy multi-constraint programming. While comprehensive, the approach's lack of detailed nutritional analysis, such as calorie content and nutritional facts, indicates a clear path for further development.

Nikahat et al. [32] introduced an intelligent system utilizing fuzzy logic to personalize meal suggestions based on health metrics like BMI and daily caloric requirements. However, their approach's omission of a multi-criteria decision-making (MCDM) framework and a comprehensive nutritional evaluation limits the effectiveness of their meal recommendations.

Raciel and their team's methodology [47], which synchronizes nutritional content with user preferences and employs the Analytic Hierarchy Process (AHP) to select meals based on macronutrient profiles, highlights the potential for integrated approaches. Nevertheless, the limitations of AHP in managing a vast array of alternatives and its uniform treatment of nutrient importance call for a more nuanced strategy, particularly in incorporating critical user considerations like budget and time.

In the exploration of meal planning, we've encountered both the potential and limitations of current research methodologies. A notable observation is the comprehensive array of dietary guidelines established by nutritionists and healthcare institutions, which are often distilled into